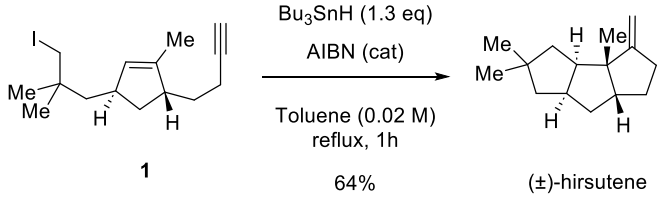
1. What are the key aspects that control reactivity in XAT reaction? (More than one answer may be correct)
   A) redox potential;
   B) bond-dissociation energy (BDE);
   C) polarizability of the C-X bond;
   D) solvent

2. What is the main side-reaction in the XAT process?
   A) Single Electron Transfer (SET);
   B) Hydrogen Atom Transfer (HAT);
   C) β-scission;
   D) β-hydride elimination

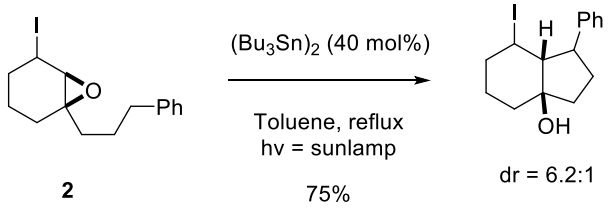
3. What type of radicals are used to initiate XAT processes?
   A) Nucleophilic;
   B) Electrophilic;
   C) Neutral

4. What are the two main methods to general radical in XAT reaction?

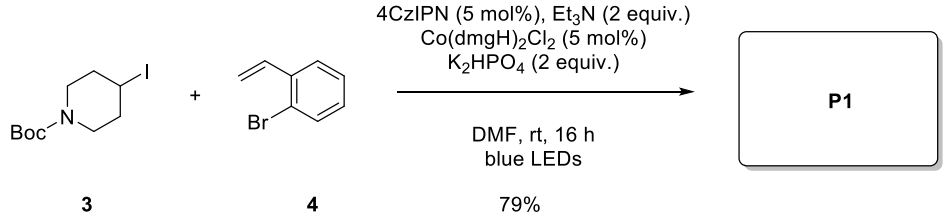
5. Dennis Curran and Donna Rakiewicz reported the total synthesis of Hirsutene (*J. Am. Chem. Soc.*, **1985**, *107*, 5, 1448-1449). Propose a mechanism of the following transformation.



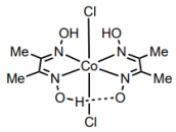

6. Viresh Rawal and Seiji Iwasa reported a photochemical rearrangement of α-iodo epoxide **2** (*Tetrahedron Lett.* **1992**, *33*, 4687-4690). Propose a mechanism.



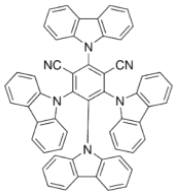

7. Leonori and Juliá developed an efficient strategy using nucleophilic α-aminoalkyl radicals and applied it in the following transformation (*Science*, **2020**, *367*, 1021-1026). Identify product **P1** and propose a mechanism of the reaction as well as an explanation of the selectivity towards alkyl iodide **3** over ArBr **4** based on the BDE provided.

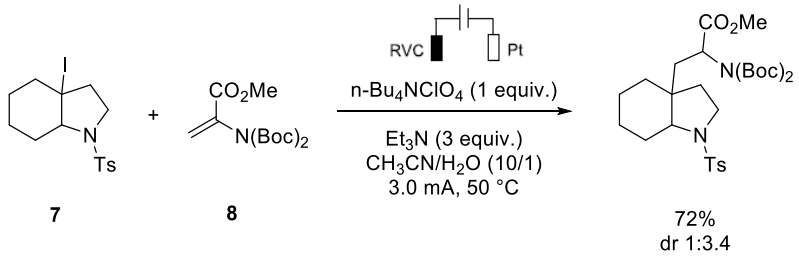



| | 3 | 4 |
|---|---|---|
| BDE (kcal.mol$^{-1}$) | 56.1 | 80.4 |




$Co(dmgH)_2Cl_2$:
Dichlorobis(dimethylglyoximato)cobalt(III)




4CzIPN:
1,2,3,5-Tetrakis(carbazol-9-yl)-4,6-dicyanobenzene

8. Xiang Sun and Ke Zheng have developed an electrochemical halogen-atom transfer (e-XAT) reaction (*Nature Commun.*, **2023**, *14*, 6825.) Propose a mechanism of the reaction.